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Object Detection using Image Dehazing : A Journey Of Visual Improvement

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***Abstract*—Object Detection in hazy conditions is very challenging as haze significantly degrades the visibility of images limits visibility especially in outdoor settings. Here we introduce an interesting method to deal with haze that is present in images. Before applying any object detection method on the hazy input image ,it is needed to be dehaze first and recognised later.For dehazing we have used the an Image Dehazing network known as All-in-One Dehazing Network (AOD-net) which is based on reformulation of atmospheric model and generates clean and clear image through a light-weight CNN and for recognition we have used the third version of famous YOLO i.e. YOLOv3 .We test our method on various real time hazy images and compare the object similarity results on hazy image as well as on dehaze image .Along with this we have compared the number of object which are recognised in hazy image and in output clear image.**

***Index Terms*—Image Dehazing , Object Detection , YOLOv3 ,CNN, AOD-net ,ASM**

I. Introduction

Artificial object recognition is gaining ground with the rise in automation and accuracy level are approaching to that of human recognition. But the majority of studies have been designed to achieve optimal performance in a clean or laboratory-like environment which ignores the most important environmental factor like weather. Due to different external conditions, the quality of many images taken in the outdoor situations are compromised.Outside Weather conditions such as fog, smoke, rain, and snow have impacted the object recognisation models significantly. Such decline in image quality caused by the weather might pose a major threat to the vision system or the camera's monitoring system's performance. To recognise object in adverse hazy situations is still a objection with many recognition techniques which performs badly on this situation so, there is a requirement to build a robust mechanism which automatically removes the haze from hazy image before going through the recognition part . We are using the atmospheric scattering model as in [1] [2] to model the images with haze. The atmospheric scattering model is given by :

$$I(x) = J(x)t(x) + A(1 - t(x)) \tag{1}$$

where , *I(x)* is Observed hazy image , *J(x)* is Original image , *A* is Global Atmospheric Lighting *t(x)* is transmission matrix. *A* also knowns as global atmospheric lighting refers to the natural light of the atmosphere across the entire scene. *t(x)* represents the amount that how much light is received by the camera from the object. It is calculated as follows:

$$t(x) = e^{-\beta d(x)} \tag{2}$$

where , β is Scattering Co-efficient and *d(x)* is distance between the object and observer . The idea behind using these equations is that light after reflected from an item is scattered by the atmosphere before it reaches the optical instrument . The quantity of light dispersed named as β is determined by atmospheric characteristics and the object's distance *d(x)* from the instrument. The remaining light is transferred towards the instrument from where it reaches the instrument . The value of *t(x)* is in the range (0, 1) since β and *d(x)* are both non-negative. The value 0 indicates that no light from the item reaches the camera (all light is dispersed), and value 1 indicates that all of the object's light reaches the camera i.e. no scattering. Greater the value of β more will be the scattering of atmospheric light. Furthermore, when the item travels further away from the camera, the amount of light dispersed increases. The light received by the camera may now be estimated using transmission values (for each pixel) and global ambient light. If *t(x)* equals 1, the camera sees the object's light properly (no scattering). If *t(x)* is 0, the camera just observes the ambient light (complete scattering). Otherwise, the camera perceives a linear interpolation between the object's brightness and the ambient light.

Object detection have made a good amount of advancement in recent years all thanks to the concept of convolution neural networks (CNN) and artificial neural networks (ANN). Object

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detection methods may be classified into two groups : one category is based on using region and another is based on using regression. RCNN [3], Fast RCNN [4], and Faster RCNN [5] are some of the object detection techniques that use region proposals. Object recognition techniques that uses regression, on the other hand, include SSD [7] and YOLO [8], among others. These approaches also build region proposal networks (RPN), however at the time of generation, these region suggestions are divided into categories. All of the approaches discussed above have proven to be effective and efficient in terms of object recognition and localization. YOLO merges labels from various datasets into a tree-like structure, although the merged labels are not mutually exclusive. Using a hierarchical design, YOLO9000 [9] improves YOLO's ability to distinguish targets exceeding 9000 categories. YOLOv3 [10] replaces the approach of estimating the cost function with multi-label classification and shows a significant improvement in discriminating small objects.

The paper is divided as follows. Section II explains about Related work. Section III describes about proposed methodology . Section IV describes about the Datasets used in this paper. Section V describes about the Evaluation and Results obtained. Section VI has Conclusion and Future Work.

II. Related Work

A. Traditional Methods

[11] is able to eliminate the haze by setting the contrast to its maximum limit. [12]By measuring the light reflected by the scene, an objectively grounded approach was presented. For clean and clear restored pictures, [13] tightened the border limit and used contextual regularisation. [14] suggested a faster approach for the automated recovery of ambient light. [15] The model parameters were supervised learnt after constructing a colour reduction prior for the foggy picture and a linear model of scene depth. In this experiment, the effective dark channel prior (DCP) is utilised.[16] to more accurately construct the transmission matrix for picture dehazing and then uses SURF algorithm for object detection which rapidly locates a point of interest that remains constant regardless of size or rotation, and is appropriate for picture matching.

B. Deep Learning Methods

CNNs have had a lot of success with computer vision problems, and they've newly been applied to haze removal. [17] used a multi-scale CNN to construct a coarse-scale transmission matrix that was then improved. [18] presented novel state-of-art model known as DehazeNet, which is trainable for medium transmission estimate. It takes a hazy image as input and produces a transmission matrix as output.Then Following the ASM , dehazeNet when combined with empirically determined global atmospheric light, yields a haze-free picture. [19] uses this DehazeNet for removing haze and then uses Faster CNN for object recognition.It accepts an input picture and presents a enclosing cube along with the probity of object being to the specific group.

Above mentioned ways have a familiar belief that for generating a clean and clear picture from a hazy image , we have to measure a correct medium transmission matrix .The ambient light is calculated separately, and the good image is rebuilt with the formula (1). AOD-Net, on the other hand, is constructed on the premise that the physical model might be formulated in a "more end-to-end" manner, with all of its parameters evaluated in a single unified model. AOD-Net will return the clean dehazed image without any intermediate technique to calculate attributes.

For object recognition both SURF algorithm and Faster CNN have some drawbacks.The SURF algorithm has a high dimension of feature descriptor, a huge amount of computation, and low matching accuracy when there is large angle of rotation and large angle of view.Faster CNN requires numerous passes for each image in order to extract all of the targets. Because of different systems function in a sequential order, the performance of the next operation is determined by the performance of the previous operations. To locate and identify the items in a photograph, this method use region proposal networks. RPNs, on the other hand, do not consider the entire picture since they only utilise the parts of it that have a high probability of containing targets.Faster CNN provides accuracy but is time consuming . Therefore to process in less time we have used YOLOv3 . The targetness score in YOLOv3 is obtained by using logistic regression. It calculates the total score for all targets within each boundary box. Because each class is assigned a logistic classifier, YOLOv3 may provide multilabel classification. YOLOv3 utilises Darknet 53 which have fifty-three levels of convolution

III. Proposed Methodology

The underlying network architecture can be splitted into two different parts : (1) Image Dehazing Network , (2) Object Detection Network. Our method is as follows : First an input hazy image is sent to object detection network i.e. YOLOv3 to recognise and classify object present in them , then the same input image is transformed to dehazed image using image dehazing network which is inspired from AOD-net . The ouput image after dehazing is again sent to object detection network to recognise and classify objects in it. Thereafter we will compare results coming from both the output images from YOLOv3.

Object detection is employed in a variety of areas of human life, including health and education. Because all of these industries are fast evolving, one-stage models must likewise evolve to meet their needs. The next advanced version of YOLO is version 3, which computes the targetness score using logistic regression. It calculates the total score for all targets within each boundry box. Because it utilises a logistic classifier for each class instead of the softmax layer used in YOLO v2, YOLO v3 may provide multilabel classification. Darknet 53 is used by YOLO v3. It has fifty-three convolutional layers. These layers are more detailed than the darknet 19 layers used in YOLO v2.

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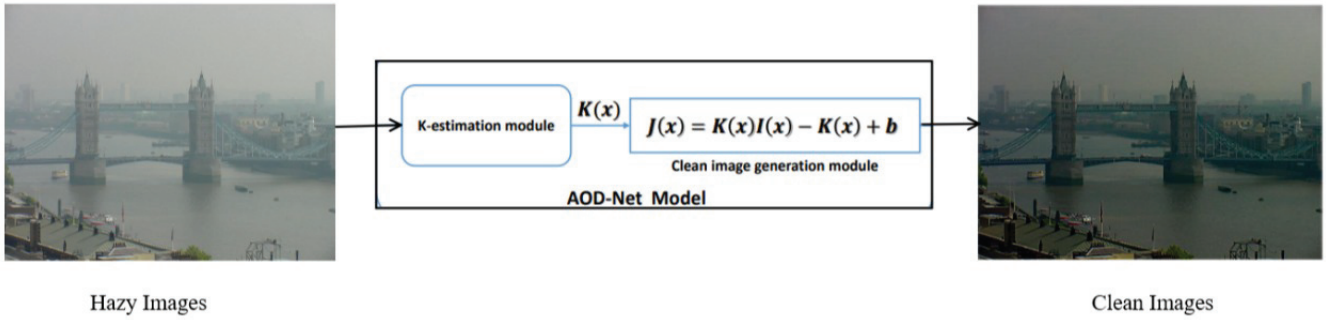

Fig. 1. AOD-net Pipeline

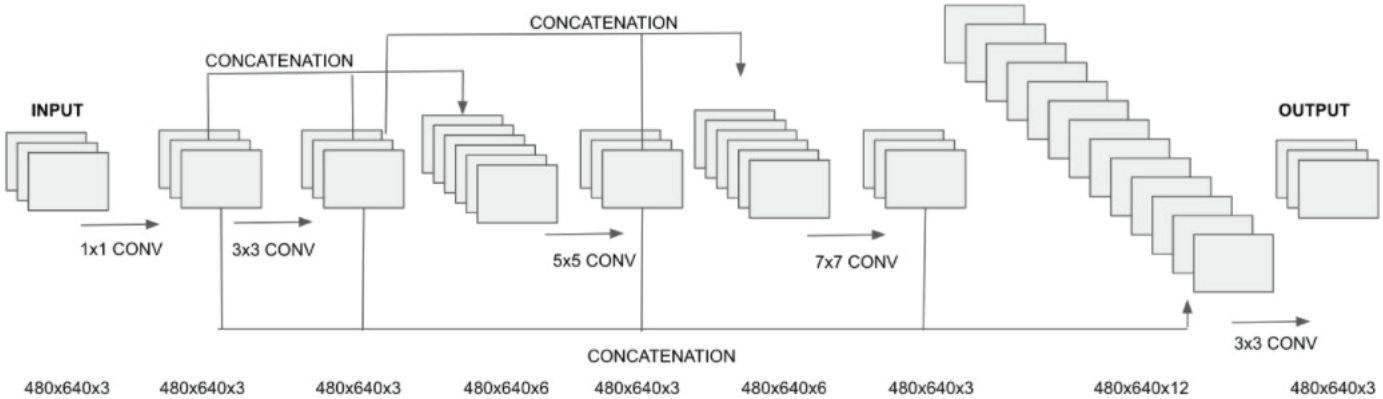

Fig. 2. K estimation module

A. Image Dehazing Network

In this section the image dehazing netork which we have used to get dehaze image i.e. AOD-net is discussed thoroughly/ First, we'll go through the modified atmospheric scattering model, which is the foundation for the AOD-Net. The AOD-Net structure is then discussed in depth.

1) Problem Formulation: Existing dehazing research relies on calculating *t(x)* and *A* separately. The primary issue here is that estimating inaccuracies can accumulates. With the following re-formulation of equations, AOD seeks to estimate both parameters in a unified manner (1):

$$J(x) = K(x)I(x) - K(x) - b \tag{3}$$

where,

$$K(x) = \frac{\frac{1}{t(x)}(I(x) - A) + (A - b)}{I(x) - 1} \tag{4}$$

t(x) and A are combined into a single variable K(x), which is altered by the input I. (x). Once we have measure the value of K(x) or K-estimated module we can generate a clean and clear dehazed image . Therefore , the primary goal now is to estimate K(x) module in such a way that the MSE between the output dehazed image and original image is as minimal as possible. This is accomplished by the use of a CNN.

2) Network Design: The used image dehazing network consists two modules.The first module is *K-estimation module* module which uses five convolutional layers to estimate the value of *K(x)* and second module is *clean and clear image generation module* which consists of several element-wise operation of multiplication and addition of layers to recover clear images using the equation (4).

The K-estimation module is heart of this network as it is responsible for indirectly estimation atmospheric light and transmission matrix. The network consists of five convolutional layers with each of layer activsted with ReLu or rectified linear function . For an image of size *W X H X 3*, *K(x)* also has size *W X H X 3*.To produce multi-scale features, each layer has three filters but a distinct kernel size. Additionally, preceding layer activations are concatenated with intermediate layer activations before convolutions to compensate for information loss during convolutions. Fig 2 states the whole architecture of network. The process initiated by sending the input 1st convolutional layer *conv1* of size 1x1 . Inspiring from [17] *concat1* layer concatanates from layers *conv1* and *conv2*. Similarly *concat2, concat3 and so on* are concatenated from *conv2*

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, *conv3* and respectively so on.The intermediate connections and this multi-scale architecture, which captures characteristics at many sizes compensate for the information loss during convolutions. By doing this we will get K-estimation value or *K(x)*.

Following the estimation of *K(x)* we can use the equation (3) to calculate of value *J(x)* which will results in generation of clean and clear image.

B. Object Detection Network

YOLOv3 model for object detection. Object identification is a basic relapse issue for YOLO, which learns enclosed box coord. and rank possibilities from an input image.

1) YOLOv3 Pipeline: It uses input set of img. with dimensions Mx416x416x3 . This image is passed to a convolution neural network where the last two dimensions of output are reduced to get an final output volume of (19, 19, 425) . every cell of 19 x 19 grid always gives 425 values .The no. of hook cubes each grid is 5, hence 425 = 5 * 85. 85 = 5 + 80, here five is (p_c, b_x, b_y, b_h, b_w) and eighty is the class count to detect.The result is a list of bounding cubes together with the classes that have been found. Six numbers are assigned to each enclosed cube (p_c, b_x, b_y, b_h, b_w, c). Each bounding box is represented by 85 values when c is expanded into an 80-dimensional vector. Finally, Intersection over Union (IOU) is worn to ignore generating overlapping cubes and Non-Max Suppression techniques which assists in overcoming the issue of recognizing an object in an image several times. It is done by choosing boxes with max. probity and ignoring low probity boxes nearby that are less than the predefined cut-off value.

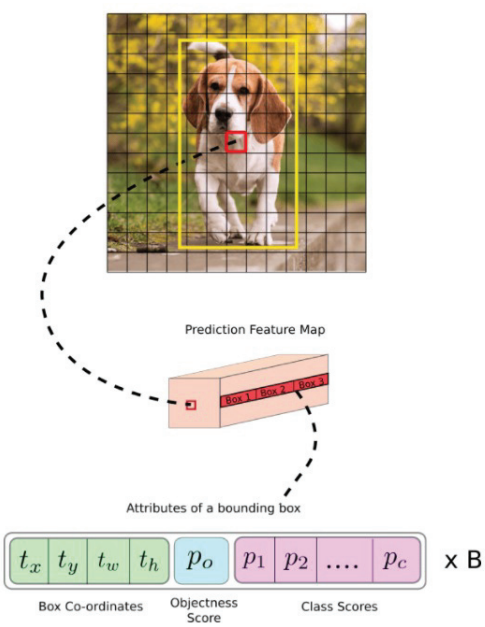

Fig. 3. Image Grid

The formulas which are written below explain how network output transform for obtaining bounding box predictions. Here, d_x and d_y are center coordinates,$d_w$is width and d_h is the height of predicted result. Top left coordinates of the grid are$m_x$and m_y. Network outputs are t_x,t_y, t_w and t_h. Anchor dimensions for the box are p_h and p_w.

$$d_x = \sigma(t_x) + m_x \tag{5}$$

$$d_y = \sigma(t_y) + m_y \tag{6}$$

$$d_w = p_w^{e^{t_w}} \tag{7}$$

$$d_h = p_h^{e^{t_h}} \tag{8}$$

Each image is divided into S x S grids, with N bounding boxes and certainty guessed for each grid. The bounding box's accuracy, as well as whether the bounding box genuinely encompasses an item, are represented by the confidence (regardless of class). Model also gives the category value for every box in every class during the training phase. To calculate the chance of each class occurring in a predicted box, you can combine the two classes.

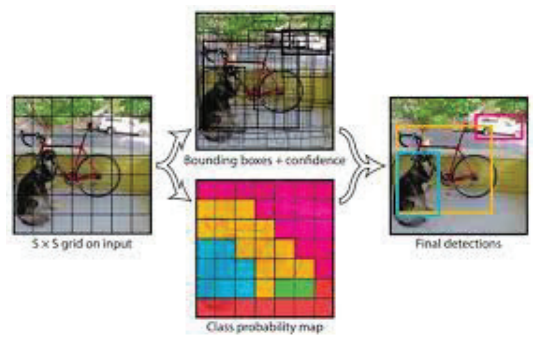

Fig. 4. Yolo Object Detection Flow

As a result, total SxSxN boxes are anticipated. However, the most of these boxes have low confidence scores, and if we establish a threshold, say 30 percent confidence, we can delete the majority of them, as seen in **Fig. 4.**.

2) YOLOv3 Network Architecture: YOLOv3 employs use of a Darknet variation that was originally trained on Imagenet and has a 53-layer network. YOLO v3 has a 106-layer fully convolution basic architecture, with 53 extra layers added on top for detection. Using 1x1 disclosure kernels on characteristic mapping of three distinct sizes at three distinct points throughout the net., this third iteration of YOLO disclosure is produced. The detection kernel has the structure 1x1x(Bx(5+C)). where B is the number of bounding boxes that a feature map cell can predict, "5" denotes the four main bounding box properties, and "1" represents the confidence level, and C is the representation of the no. of groups. The kernel size of YOLO v3 trained on COCO is 1 x 1 x 255 since B = 3 and C = 80. Assume we have a 416 × 416 input picture. YOLO v3 model gives prediction at three different scales, which are calculated by downsampling the input image's dimensions by 32 pixels, 16 pixels, and 8 pixels, respectively.

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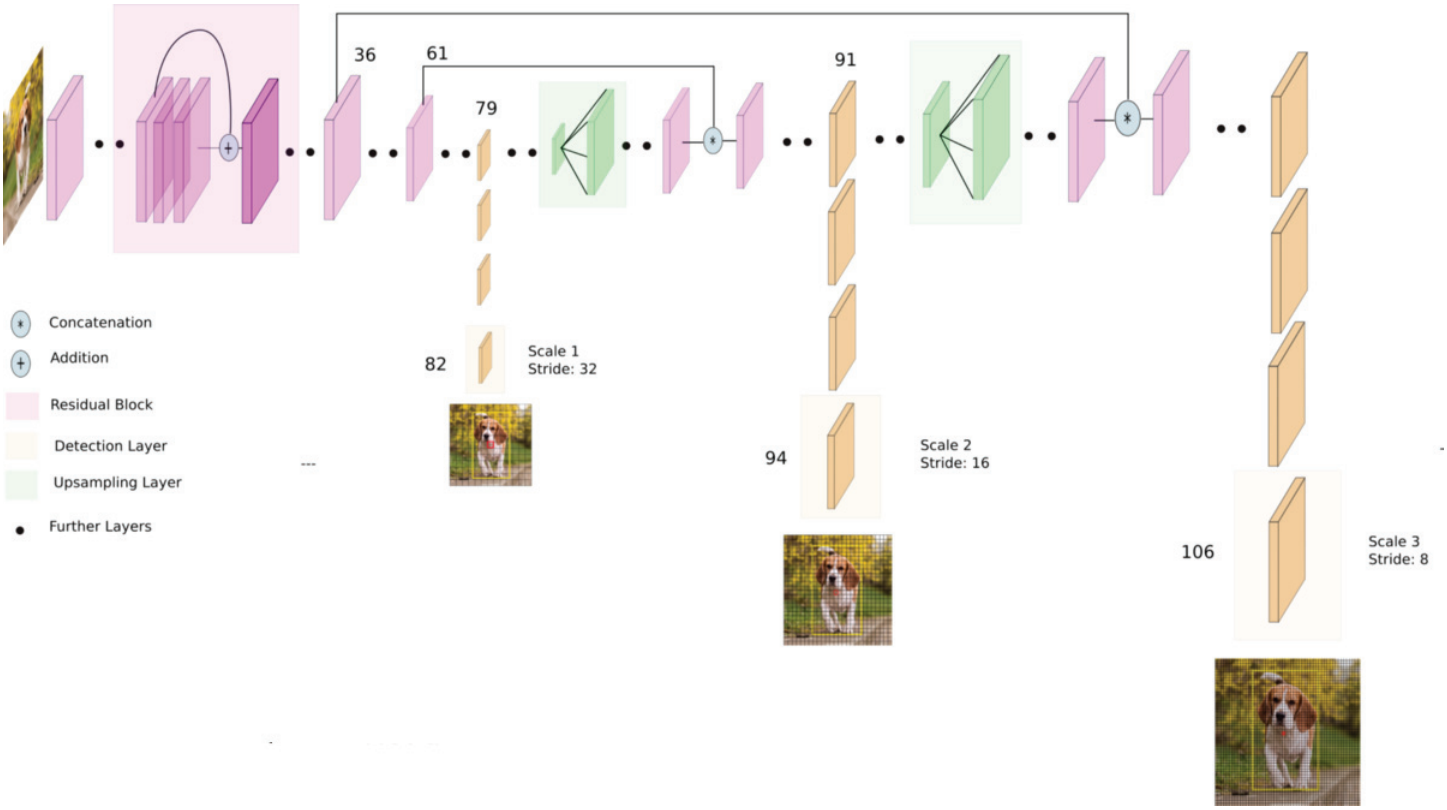

Fig. 5. **The Architecture**. YOLO v3 is based on a Darknet variation that used to have a 53-layer net. trained on Imagenet. In order for recognition, fifty three successive outlets are put on top, giving YOLO v3 a hundred and six layer deep connected underlying architecture.

Initial detection is performed by the 82nd layer. The network down samples the 1st 81 layers of the picture, with the 81st layer having a stride of thirty two. The final feature map will be 13 × 13 if we have a 416 x 416 image. We perform one detection using the 1 x 1 detection kernel, resulting in a 13 x 13 x 255 detection feature map. The feature map from layer 79 is subjected to a few convolution layers before being up sampled by two times to 26 x 26 dimensions. After that, the layer 61 characteristic map is depth combined with this characteristic map. After that, the combined feature maps are fed into numerous 1 x 1 convolution layers, which fuse the values from the former layer (61). The 94th layer then performs the 2nd detection, giving in a disclosure characteristic map of 26x26x255. Before being depth combined with a feature map from layer 36, the feature map from layer 91 is processed to a few number of convolution layers. With effect from the previous layer, a few 1x1 convolution layers are added to include the info (36). We create the final of three feature maps at the 106th layer, providing a 52 x 52 x 255 feature map.

IV. DATASETS

Artificially hazed pictures are used for the training phase of the Net. For this purpose we are using NYU2 Depth Database [20] which stores depth information of each and every pixel.It contains hazy images in outdoor, urban , wild, indoor, day and night environments . Images were artificially hazed with several values of β and A which are selected randomly from uniform distribution based on equation (1) .They used a fog system to generate the haze, then recorded pictures of the identical subject at distinct degrees of haze (a overall of ten) ranging from zero haze to entirely wrapped in haze. Moreover for about 1500 groundtruth images there were 27K hazed images which are obtained.The model was trained on a total of ten epochs. Weight decay was fixed to 0.0001 and weights were set with a regular Gaussian distribution. The gradient norms were set at 0.1. We employed a learning rate of 0.0001 and Adam Optimizer to speed up the gradient descent process by taking into consideration the 'exponentially weighted avg. of the gradients in our implementation.

For training YOLO v3 model we have used COCO dataset [21] which contains contains challenging, high-quality visual datasets for computer vision, mostly state-of-the-art neural networks.The COCO dataset is well-labeled, provides data to train supervised models that are able to identify the common objects in the dataset. Moreover COCO dataset consists of 80 different object categories some of which are people , car , dog , trucker etc.

V. EVALUATION AND RESULTS

On doing analysis based on various metrics ,and going through some tabular data ,that is created based on our observation and results of the output that we have received from our model,we are in a position to discuss the results that

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Fig. 6. Images with haze levels of 1 to 10

we have been able to observe, the hazy images that we used as input shown some interesting results when dehazed ,the results varied depending on the levels of haziness in the image, for results one can refer to the image comparison in *fig.7*.

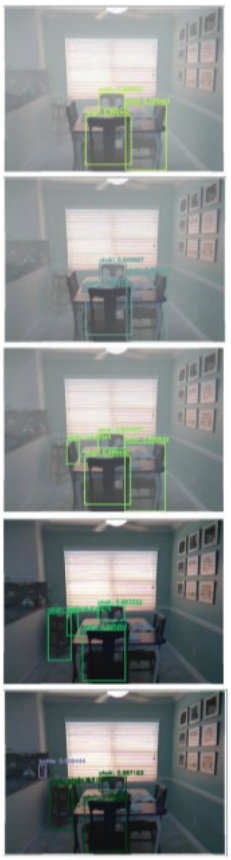

Fig. 7. Number of objects recognised at different hazy levels

From the table 1 and fig.7 we can see a glimpse of what wonders can dehazing model can do to our object Detection model, the results of object recognition get better and are more visible with decreasing haze level. As one can deduce from the observations in the table.1 , that with increasing level of haze in the input image, can make object Detection algorithm less effective . As one can see of haze level 5, the object Detection algorithm can only detect 3 object in an image with haze, as compared to Dehazed Image,where number of object detected are 5, trends follow in case , when we decrease the haze level to 4, that is object Detection algorithm can only detect 3 object in an image with haze, as compared to Dehazed Image,where number of object detected are 5. But as soon as we reach haze level 3 ,we see some improvement as number of detected objects for hazy image is increased from 3 to 4, this results is seen again for haze level 2,but as we reach haze level 1 ,i.e the level at which the haze is negligible or image is dehazed , there we see most number of objects detected so far i.e 5, which is equals to number of object detected for dehazed image, from our observation ,we can conclude that there is significant improvement in object detection functionality of our algorithm as haze decreases. So therefore in dehazed image number of objects recognised is more as compared to that of dehazed image.

TABLE I
NUMBER OF OBJECTS DETECTED AT VARIOUS HAZE LEVEL.

Haze Level	Objects Detected in hazy image	Objects detected in Dehazed Image
5	3	5
4	3	5
3	4	5
2	4	5
1	5	5

In Fig. 8 , we are witnessing the real usage of image-dehazing in object-detection using real world examples, one can easily decipher the riddle of how new objects are detected in Dehazed image as we mention and discuss the number associated with the images in Table 2. So let's begin with Image 1, here we can see that in hazy image ,our algorithm successfully detects the object ,i.e a bed with a accuracy of about 97% ,whereas the object-detection Algorithm detects the same object with an accuracy of about 99% ,that is nearly 2%

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Fig. 8. Object Detection on hazy Image (left-side) and respective Dehaze image (right-side)

TABLE II

S.No	Objects recognized in hazy image	Similarity percentage of each object	Object recognized in dehazed image	Similarity percentage of each object	Diff in accuracy
1	1	Bed 97%	1	Bed 99%	2%
2	3	Chair1 55% Chair2 65% Book 74% Chair3 (Not detected) Chair4 (Not detected) Chair5 (Not detected)	6	Chair1 92% Chair2 92% Book 72% Chair3 64% Chair4 80% Chair5 88%	47% 37% 2% 64% 80% 88%

better as compared to the hazy image. Similarly for Image 2 where we can an image with clear multiple objects,our object-detection Algorithm detects 3 objects in case of hazy image, the objects are 2 chairs with accuracy of 65% for near one and 51% for farther one , and a book with an accuracy of 74% , whereas for dehazed image ,the algorithm detects 6 objects including those which are previously detected in hazy image. let's compare results for this observation those two chairs which were detected before are now detected with accuracy of 92% each ,i.e an improvement of about 27% and 41% respectively. also book is also detected with an accuracy of 72%, this abnormal behaviour ca be explained by the fact that is haze is decreased the object appears less like a book and more like bunch of paper or napkins ,which leads to decrease of nearly 2% in its identification.Other than that the algorithm also detected 4 new chairs in the dehazed image with an accuracy of about 92% ,80%,84% and 88% clockwise starting from bottom one. which leads us to the conclusion that our object-detection algorithm (YOLO V3) performs better if we dehaze a hazy image and hence gives better results for real world scenarios.

VI. Conclusion and Future Work

In conclusion of this paper, we have successfully found a way to improve the extent of object-detection in hazy images also we have shown the quantitative results so obtained, by comparing the results received from an image of hazy environment and results received from dehazed version of the same image, the results were both surprising and insightful. As we come across many scenarios where the resultant image has less accuracy as compared to that of hazy Environment ,that's because Dehazing causes images to darken, which makes recognition more difficult. This has left a unfavourable brunt on the picture's small element or objects.

Actually our main use case is using this in security cameras. We are planning to find a way of finding haze estimation in an image which depending on the threshold value will tell us whether the given image is needed to dehaze or not. This approach will make dehazing network to decide whether to dehaze an image or not when there is a modest amount of haze

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it will directly undergo recognition reducing the computational time of removing haze.Also we will try to include videos haze removal which will reduce the haziness in each an every frame of the video .

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